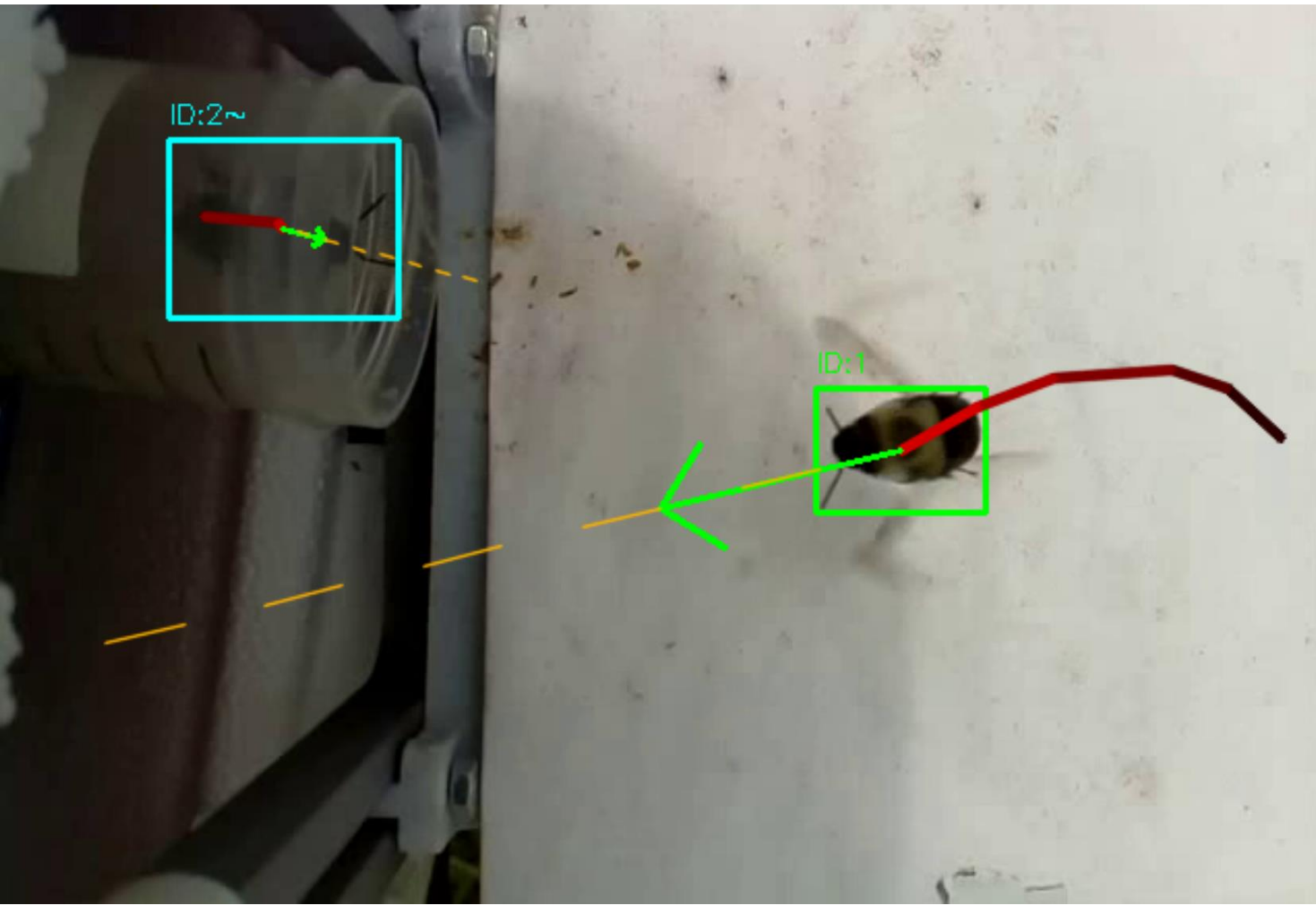


MOTIVATION

Bees pollinate a large share of the world's food crops, yet populations have declined for decades. Hive-entrance video captures exactly what researchers need: foraging rate, traffic patterns, stress response. But raw footage is useless without labels; training a reliable detector has traditionally meant hand-drawing thousands of bounding boxes per hive.

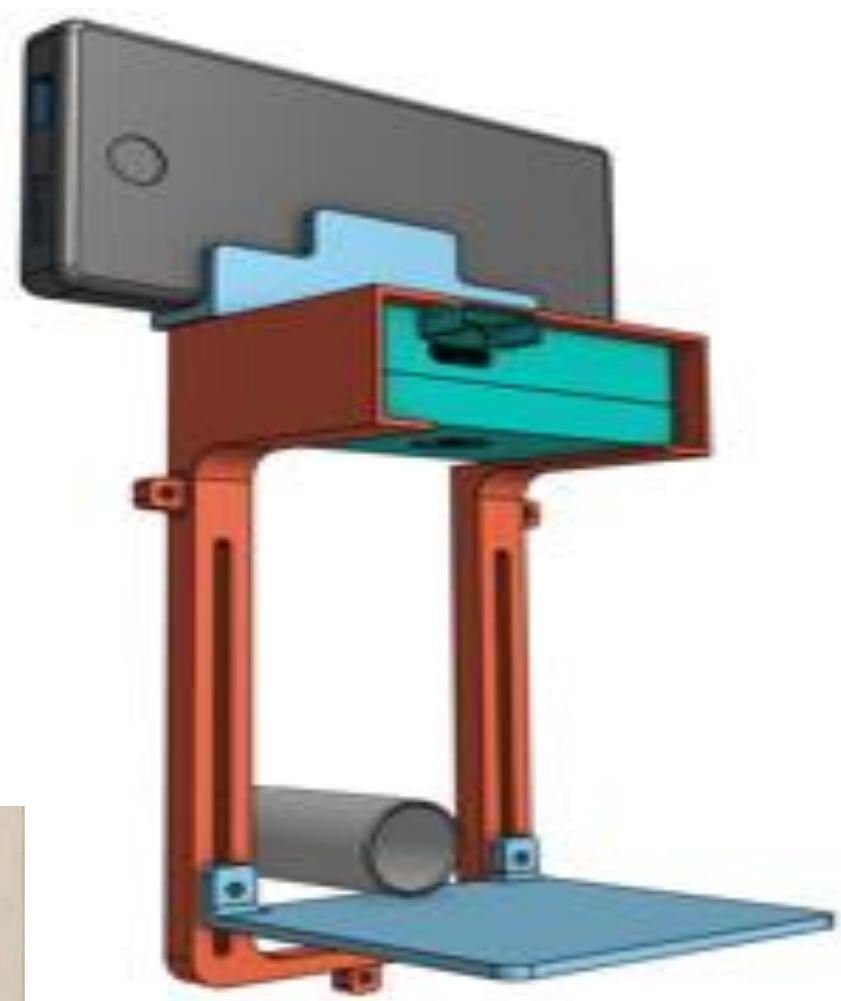
BeeTracker collapses that pipeline into an iterative loop of auto-label, review, and fine-tune, so one researcher can go from raw video to trajectory data in an afternoon.



Multi Bee moment. Left track is velocity reacquire, right track is a full lock

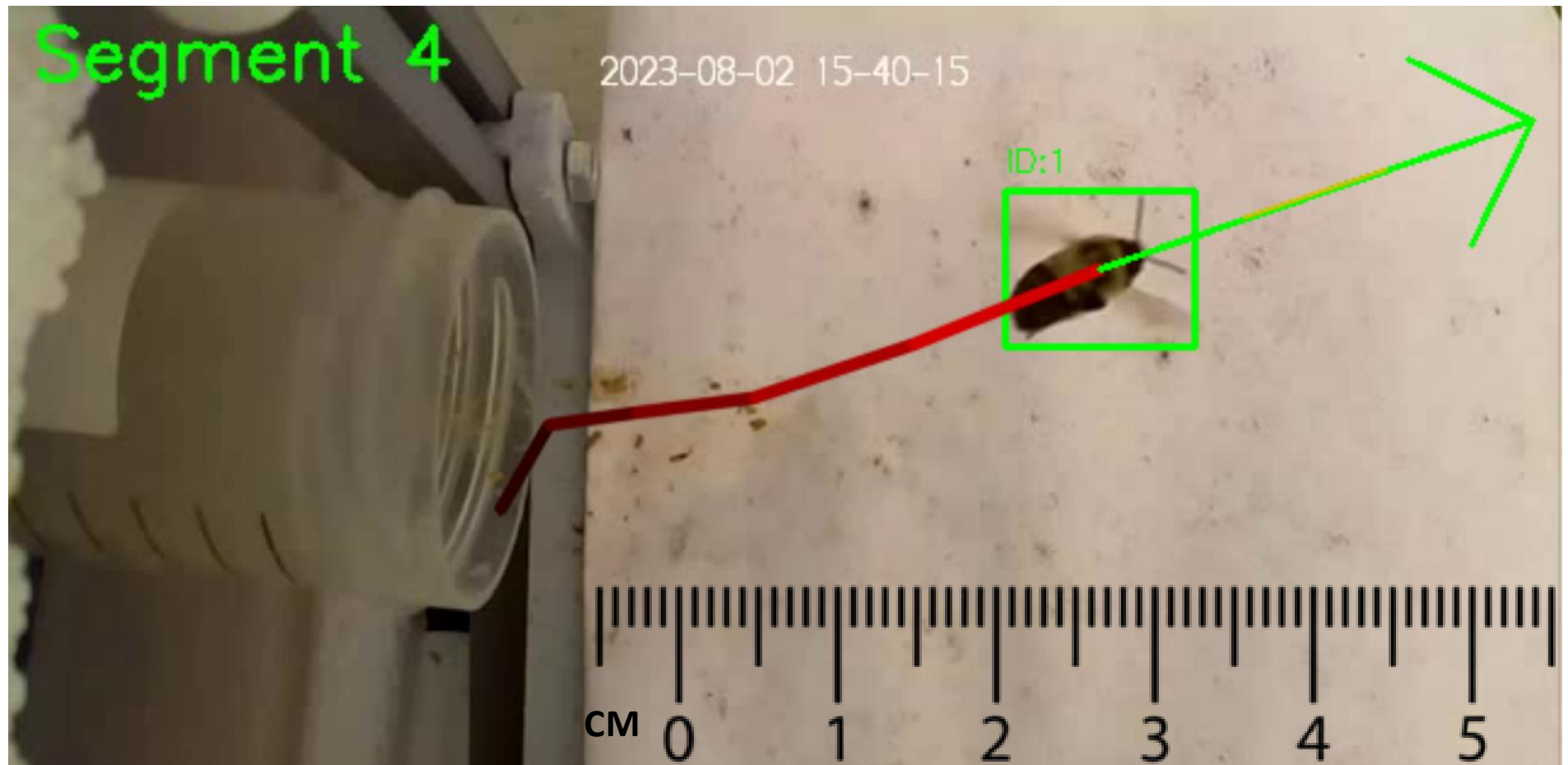
DATA & COLLECTION

Raw input: hive entrance video, recorded at an outdoor apiary. The whole system is a camera, a stand, and a mesh enclosure.



Field enclosure (top-left), Camera mount (top-right), Example of user marked hive entrance (bottom-right)

DETECTION & TRACKING

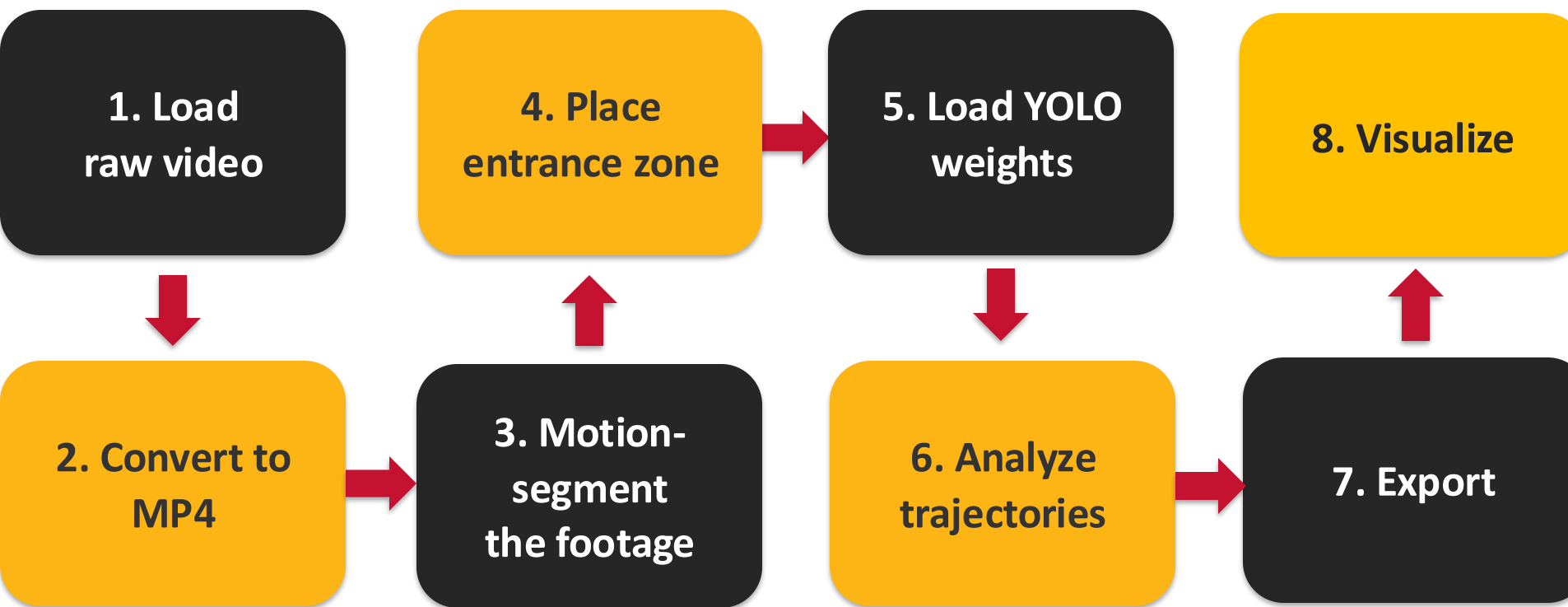


Live frame: YOLO detections plus velocity-vector trajectory overlay.

Each frame passes through a fine-tuned **YOLO** detector. A ByteTrack layer links detections into persistent tracks, and a **velocity-vector** corrector recovers tracks missed between frames. Completed tracks are classified as entry, exit, or neither against a user-placed entrance zone.

RUNTIME WORKFLOW

End-to-end pipeline on a new hive video, from raw file to exported trajectories. The 'convert' step normalizes MOV, MKV, H.264, and AVI inputs to MP4 so OpenCV decodes frames deterministically; the rest runs inside the desktop app.



CLASSIFICATION SCORE

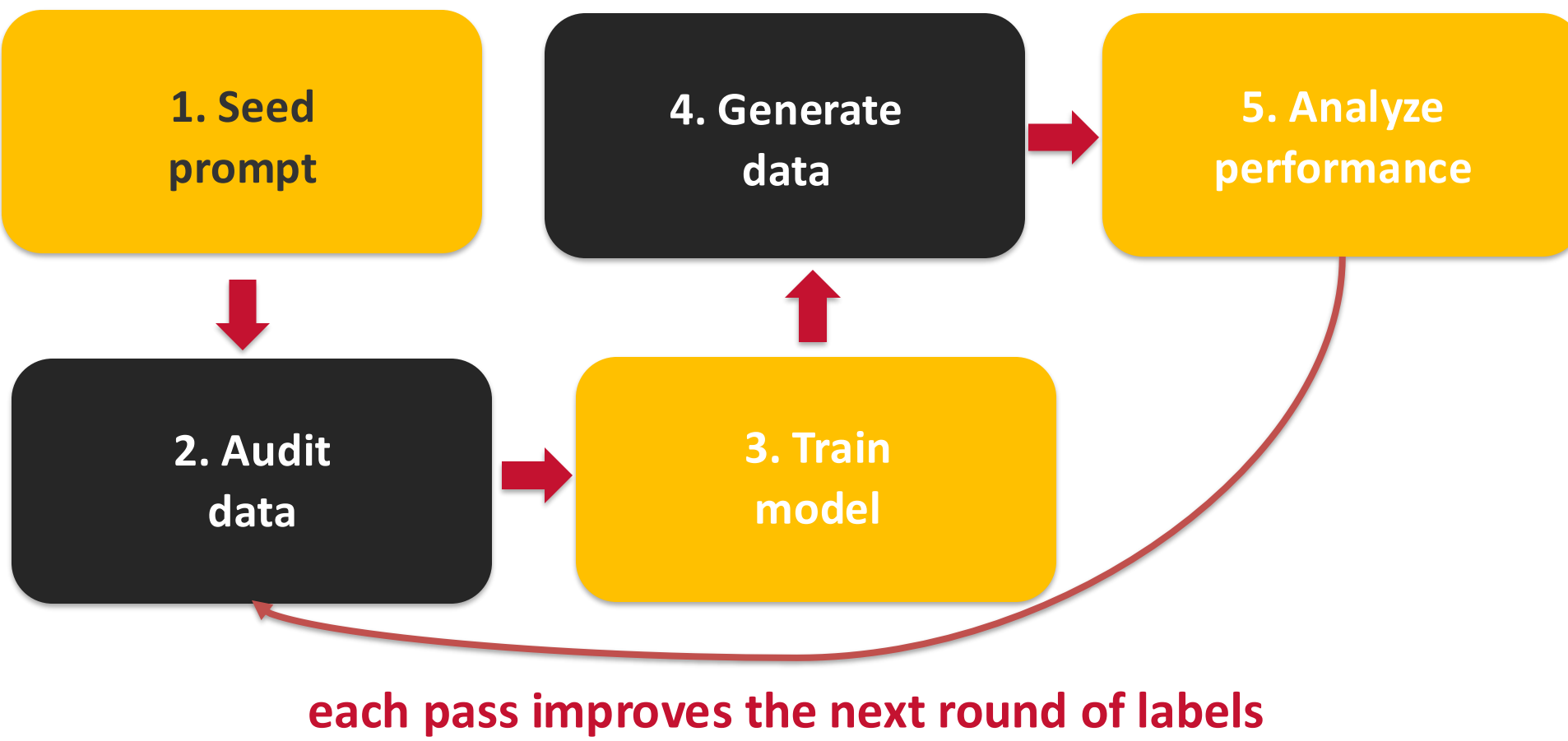
$$\text{score} = 0.35 \cdot \text{displacement} + 0.35 \cdot \text{velocity-alignment} + 0.20 \cdot \text{terminal-proximity} + 0.10 \cdot \text{closest-approach}$$

|score| > 0.15 → entry (+) or exit (-); otherwise "unknown".

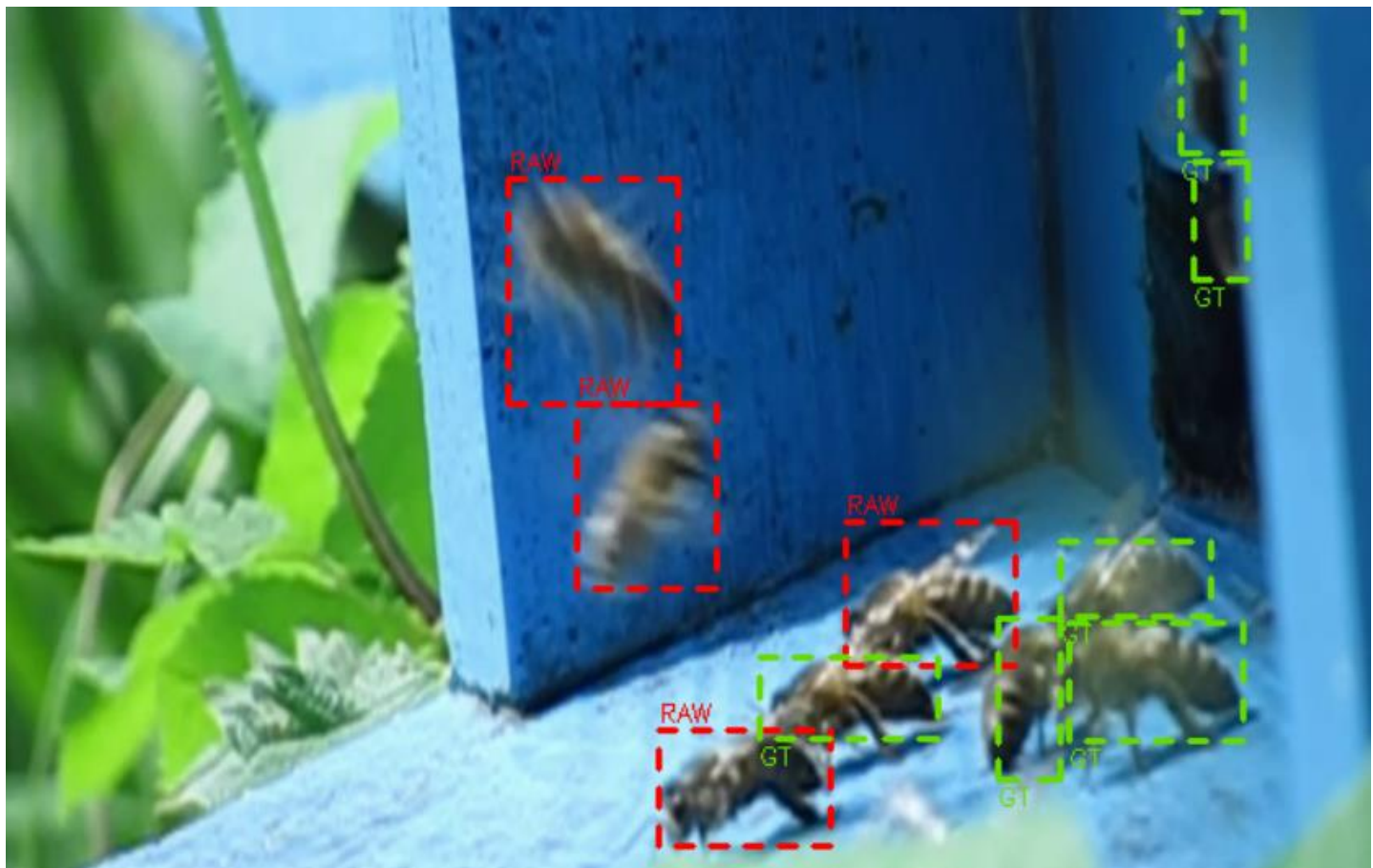
Gate: bee must dwell near the entrance for ≥3 frames.

AUTOLABEL LOOP

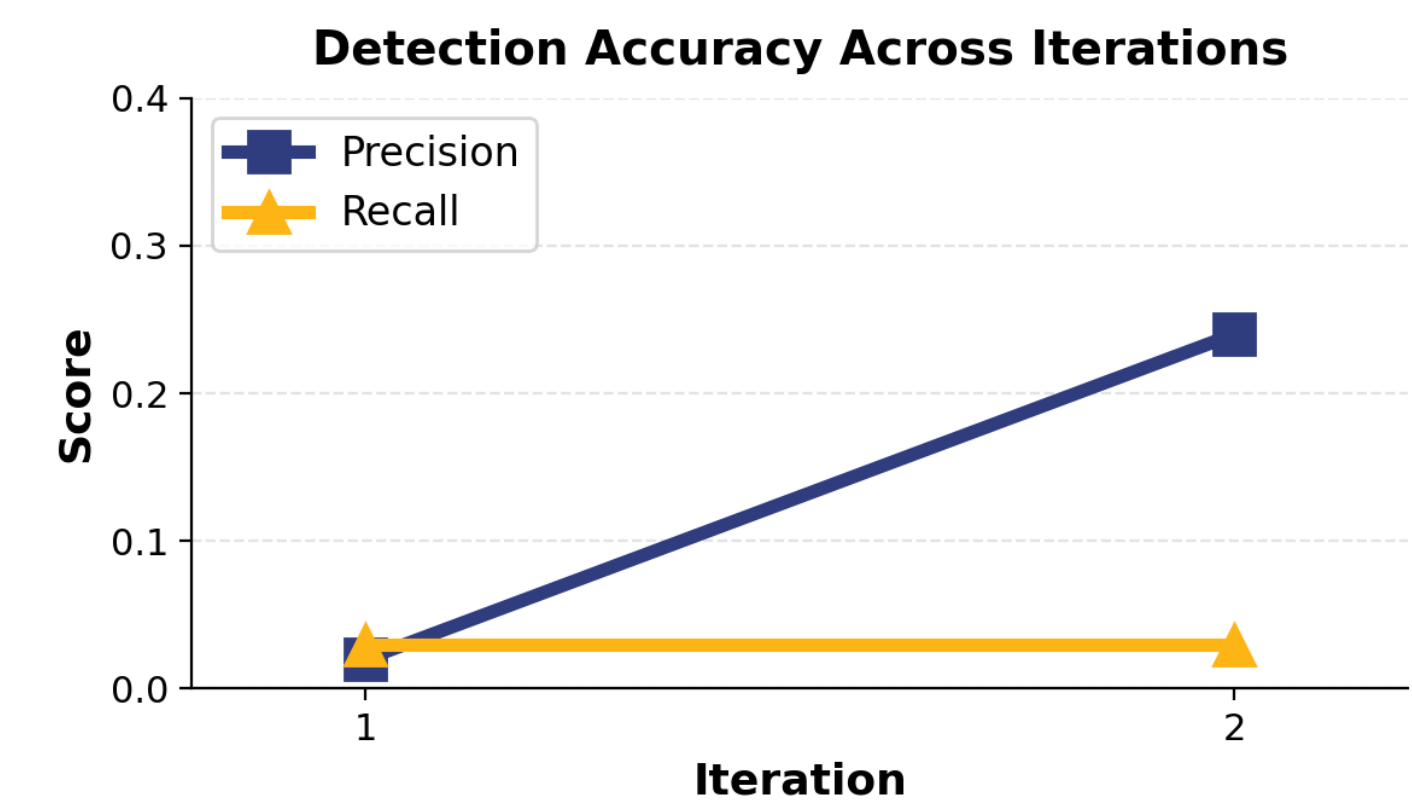
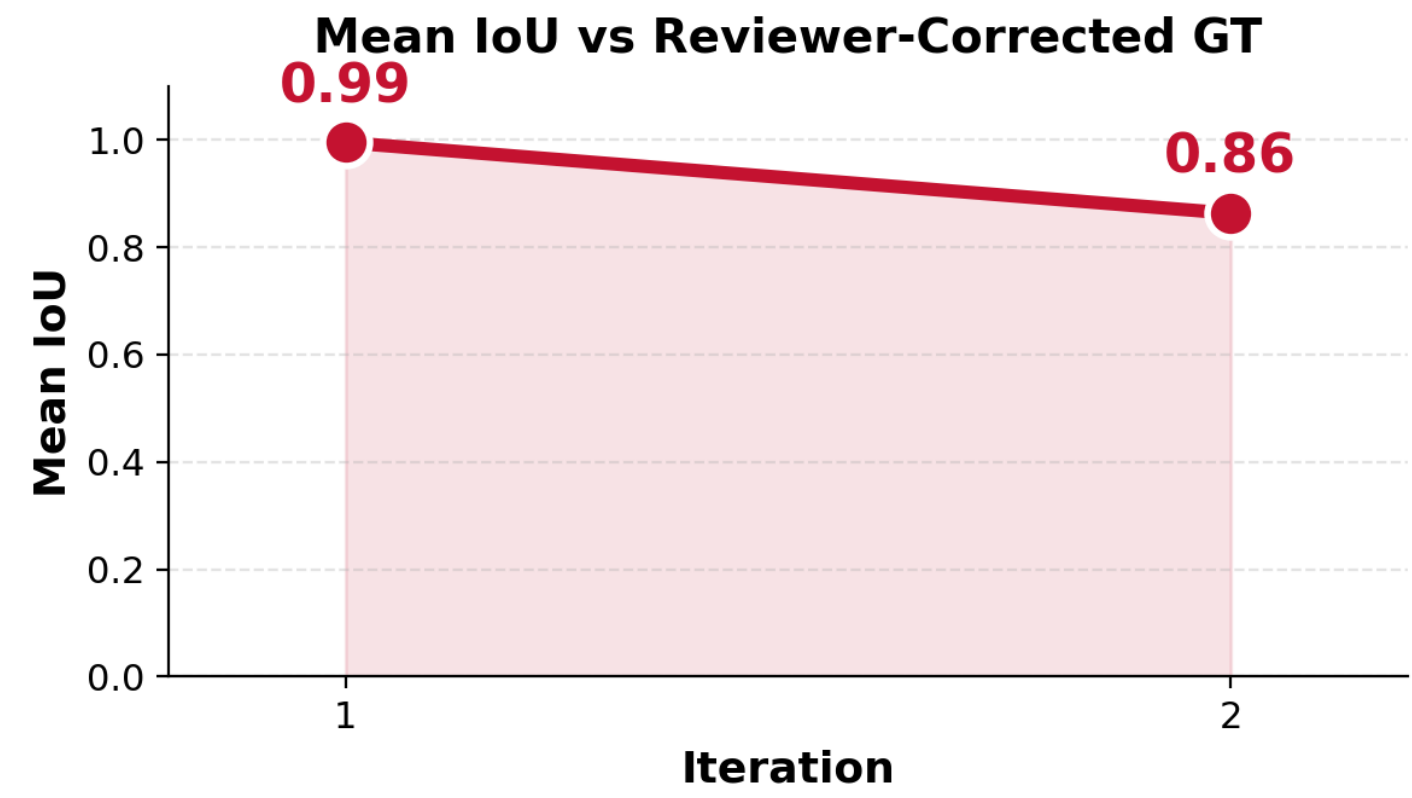
An iterative loop that turns a text prompt into a fine-tuned bee detector.



AUTOLABEL PERFORMANCE

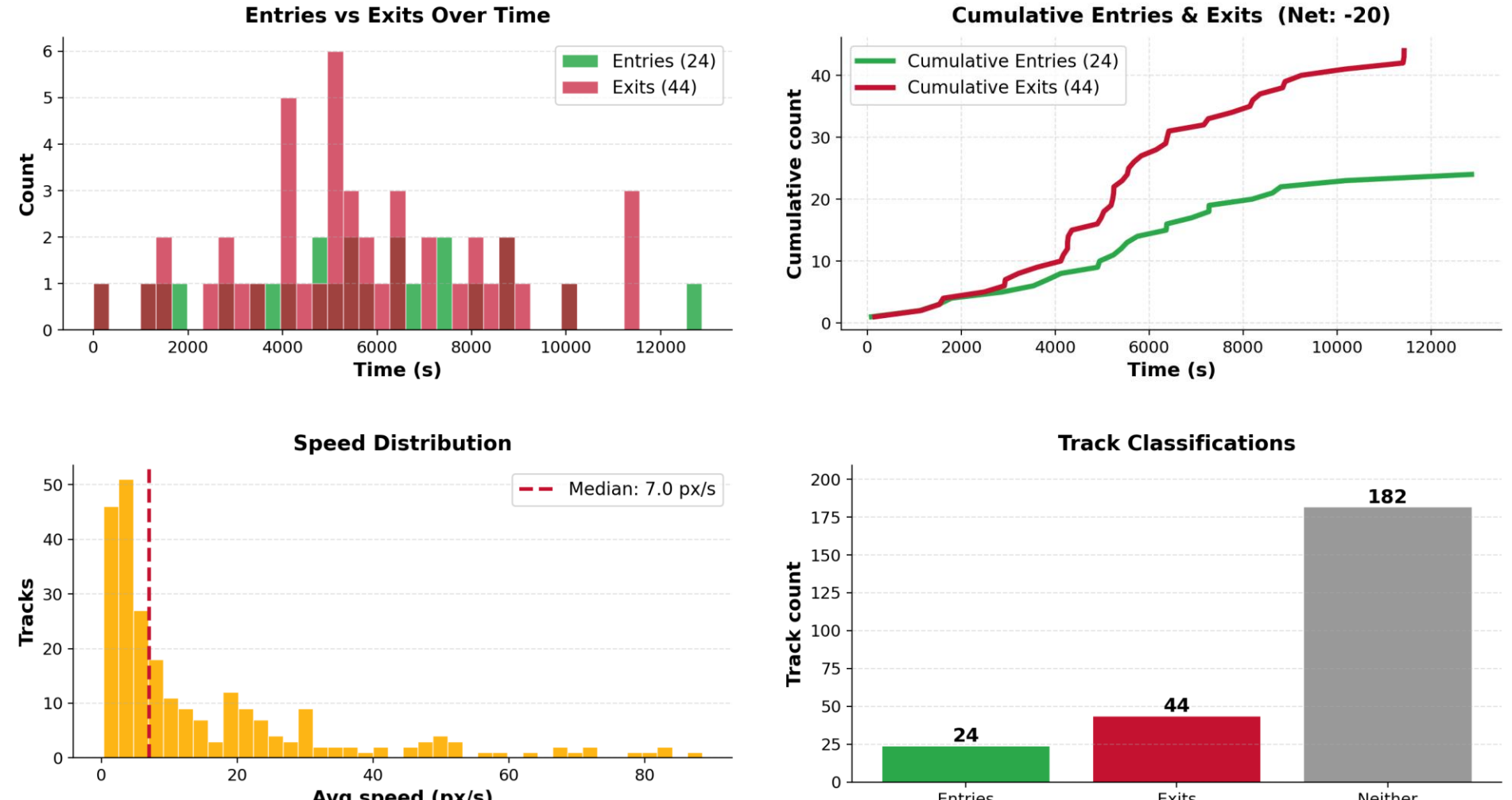


Raw YOLO predictions (red dashed) vs reviewer-corrected ground truth (green dashed).



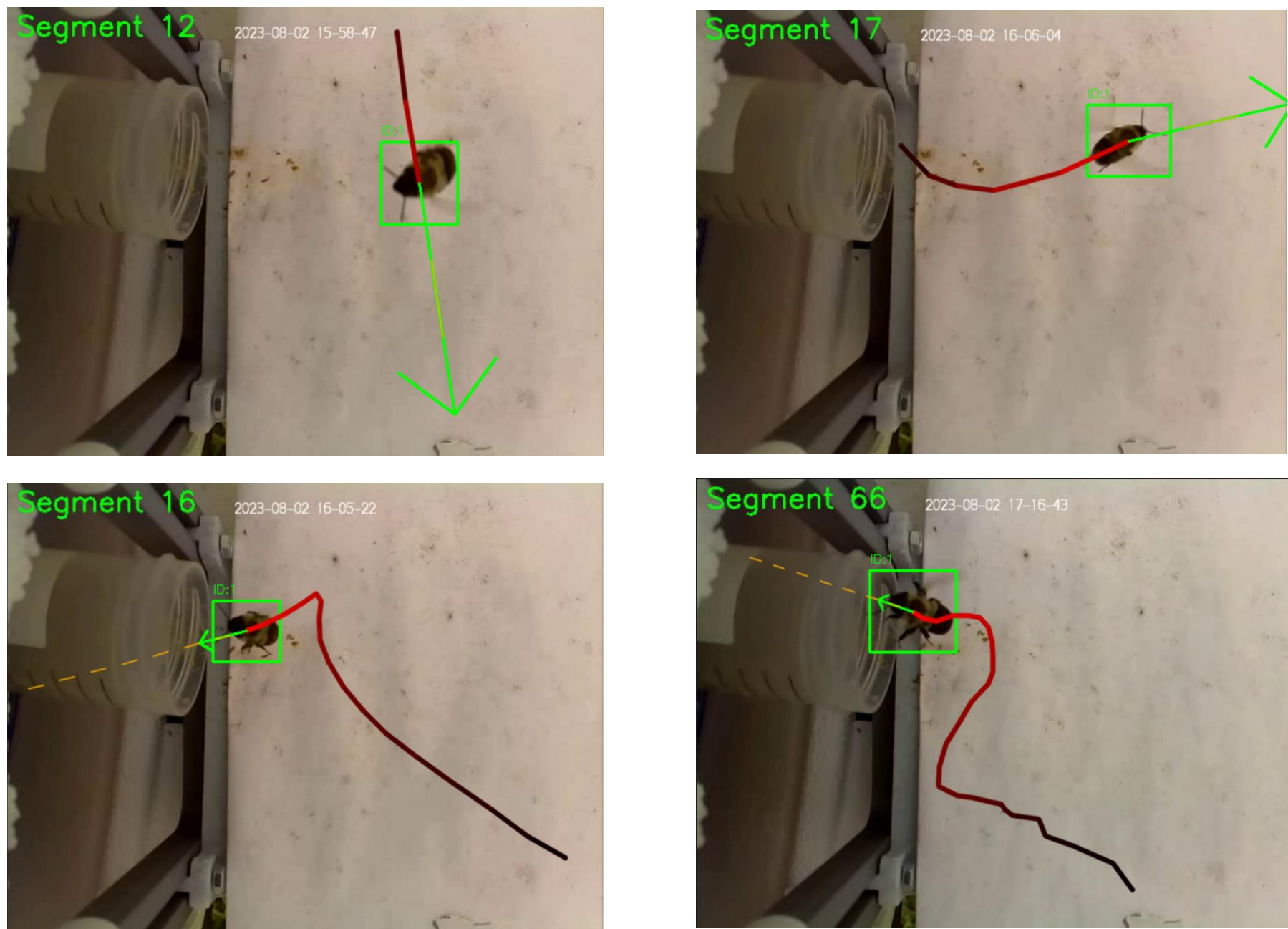
Mean IoU holds at ~0.86 after the 2nd iteration; predicted boxes align tightly with reviewer-corrected ground truth. Precision rises as YOLO learns bee-specific cues.

TRAJECTORY VISUALIZATION



24 entries, 44 exits detected across a 165-minute session, condensed to 143 navigable motion segments. Final tracks export to CSV or JSON; a manual override UI handles edge cases.

SAMPLE TRAJECTORIES



Entry, exit, and orientation-flight paths overlaid on source frames.

FUTURE USE CASES

The pipeline generalizes past bees. Anything that crosses a boundary is fair game: ants at a nest, rats in a lab study, people walking through a doorway. Load a video, sketch an entrance, label a handful of frames, and the rest runs itself.

ACKNOWLEDGMENTS

Thanks to Guadalupe Bernal for starting this research and providing every hive video used to test and tune the pipeline; none of the results above would exist without her. Thanks also to Zeynep Temel for advising this project and giving me the opportunity.